PHY-201 Final Examination (Draft) - Fall 2026

SECTION A: MECHANICS

1. Calculate the trajectory of a projectile launched at 45 degrees with an initial speed of 20 m/s and find its range. (10 marks)

2. A 5 kg block is pushed across a frictionless surface with a force of 20 N. Calculate its acceleration. (5 marks)

3. Explain elastic collisions. (10 marks)

(a) Derive the equation for conservation of kinetic energy in a one-dimensional elastic collision. (5 marks)

(b) Calculate the velocities of two equal masses after a head-on elastic collision. (5 marks)

SECTION B: THERMODYNAMICS

4. State the second law of thermodynamics and explain the working of a Carnot heat engine. (15 marks)

5. Design an experiment to measure the specific heat of an unknown metal and evaluate its main sources of error. (10 marks)